

AIPL Runtime Feature Matrix

7 implementations compared

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2026-05-18: Phase O complete. OCaml AIPL now matches Python (annotated) on all 18 Phase C-E2 + AIPL v2 Distributed features (CE-1..CE-9 + DR-1..DR-9). JS-OCaml inherits these through the OCaml backend. See `docs/OCAML_PORT_ROADMAP.md` and the Phase O summary at the end of §Phase C-E2 and §AIPL v2 Distributed below.

Overview

AIPL (Actor-based Intelligent Parallel Language, formerly AIPL) has grown into a **seven-runtime project**. This document provides a side-by-side comparison of every language and infrastructure feature across all implementations, grounded in source surveys.

Legend. ✓ = full support, △ = partial, ✕ = not supported, – = not applicable.

	Short name	Source
The seven runtimes.	Python (annotated)	<code>src/python-aipl/</code>
	Python (inferred)	<code>src/python-aipl-inferred/</code>
	OCaml	<code>src/*.ml, _build/.../repl_thread.exe</code>
	JS-OCaml (server)	<code>src/app.js, console_server.js, ide.js</code> + OCaml <code>web_gateway</code>
	JS-Browser (serverless)	<code>src/browser-abcl/</code>
	JS-Node (server)	<code>src/node-aipl-server/</code> (Node HTTP server wrapping browser-abcl)
	C (aipl2c)	<code>src/aipl2c.ml</code> codegen + <code>src/abcl_gui_runtime.c</code>

Column abbreviations in matrices below. **Py-A** = Python annotated, **Py-I** = Python inferred, **OCaml**, **JS-O** = JS via OCaml backend, **JS-B** = JS in browser, **JS-N** = JS in Node server, **C** = aipl2c codegen.

Inheritance relationships.

- **JS-OCaml** is a thin HTTP client over the OCaml backend, so its feature set inherits OCaml's (plus the new `/api/typecheck` endpoint).
- **JS-Node** wraps the same browser-abcl parser / type-checker / interpreter inside a Node.js HTTP server. Runtime feature set therefore mirrors JS-Browser.

- **Python (inferred)** imports parser, interpreter, AI, remote, and dashboard from `../python-aipl/`. All *runtime* features match Python (annotated); only the *static type-checker* layer differs (full Hindley-Milner instead of Phase 11+ annotation-driven).

Sample programs

Number of `.abcl` sample programs reachable by each runtime (core directory + AI integration + remote-actor demos):

	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
core samples	33	33	57	57	5	5	57
AI integration samples	11	11	8	8	0	0	8
remote-actor samples	8	8	1	1	0	0	1
total	52	52	66	66	5	5	66

Cross-cutting / not tied to a specific runtime:

- `aipl-self-host/`: 48 AIPL-in-AIPL self-host programs
- `docker/cross/samples/`: 4 cross-language interop samples

Note that the C runtime processes the same `.abcl` files as OCaml via `aipl2c`; of the 66 reachable samples, 48 currently pass the `aipl2c` smoke test (the remaining 9 in `abclc/` have pre-existing type ambiguities surfaced by our hard-fail policy).

What each sample exercises

Samples are grouped by the language or infrastructure feature they demonstrate. “Where” points to the source directory; “Target” lists the runtime(s) and codegen targets that actually execute the sample.

Target abbreviations: **Py** = python-aipl / python-aipl-inferred; **OCaml** = OCaml REPL (`abclc`); **C** = `aipl2c` → C+pthread; **SDL2** = `aipl2c` → C+SDL2 GUI binary; **Xinu** = `aipl2c --xinu` → Xinu-flavoured C; **Py-gen** = `aipl2c --python` → stand-alone Python; **Pony** = `aipl2c --pony` → Pony actor source; **Erlang** = `aipl2c --erlang` → Erlang module; **Go** = `aipl2c --go` → Go source; **Prolog** = `aipl2c --prolog` → SWI-Prolog `.pl`; **JS-B** = browser-abcl; **JS-N** = node-aipl-server.

Feature category	Sample(s)	Where	What it checks	Target
Basic actor model now/future/await	Hello, Counter, PingPong NowFuture, CooperativeNowFuture, NowFutureDemo	py-aipl + abclc py-aipl + samples-ai + abclc/ai-samples	actor creation, <code>send</code> , field state, <code>self-send</code> three message-passing forms in one program	Py, OCaml, C, JS-B, JS-N Py, OCaml, C, JS-B, JS-N
Bounded mailboxes / select	BoundedBuffer, bounded_buffer{,_visual}	py-aipl + abclc + browser-abcl	producer/consumer, selective receive, visualisation	Py, OCaml, C, JS-B, JS-N
Dining philosophers	Philosophers, philosophers{-1,-2}, Philosophers5{,_debug,_trace,Gui,Py,Xinu}	py-aipl + abclc + browser-abcl	fork as actor, deadlock-free serialisation, SDL/Python/Xinu variants	Py, OCaml, C, SDL2, Py-gen, Xinu, JS-B, JS-N
become (class swap) select / selective recv	become, bbecome Channels, Channels2	abclc py-aipl/samples	runtime actor-class replacement typed channels, worker pool with request/result	Py, OCaml Py
Method injection	MethodPatch	py-aipl/samples + abclc/	<code>add_method/remove_method</code> runtime patching	Py, OCaml

Feature category	Sample(s)	Where	What it checks	Target
Dynamic compile	Dynamic, DynamicWorkerPool	py-aipl/samples + abclcl/	<code>compile()</code> builtin → runtime class generation	Py, OCaml
Top-level functions	Functions, Signatures	py-aipl/samples + abclcl/	user functions, overload signatures, <code>typeof</code>	Py, OCaml
Records	Records	py-aipl/samples + abclcl/	<code>{a:int, b:string}</code> literal, field access, structural <code>typeof</code>	Py, OCaml
Tuples	Tuples	py-aipl/samples + abclcl/	positional, immutable, mixed slot types, nesting	Py, OCaml
Arrays	Arrays, MultiDimArrays	py-aipl/samples	static-sized, multi-dim, dynamic sizes	Py
Gradual type checker	Typecheck, Typecheck11{b,c,de}	py-aipl/samples	Phase 11 → 11e: literals, call-site validation, unions/generics, narrowing, length-tagged arrays	Py
Phase 11 typed counter	Phase11_TypedCounter	abclcl	typed annotations / generics / <code>typeof</code> narrowing	OCaml
Phase 12 effects	Effects, Phase12_EffectsLog	py-aipl + abclcl	capability-based <code>!{fs,ai,net,mut}</code> system	Py (static), OCaml (sample)
Phase 13 channels	Phase13_Channels	abclcl	CSP channels (design doc + runtime in Py)	Py (runtime), OCaml (sample)
Phase 14 linear types	Linear, Linear2, Phase14_Linear	py-aipl + abclcl	use-after-move; DB transaction with linear handle	Py (static), OCaml (sample)
Phase 15 owned fields	Owned, Owned_violations, Phase15_Owned	py-aipl + abclcl	<code>pub</code> field visibility, intentional violations	Py (static), OCaml (sample)
Phase 16 transient cast	Transient, Transient_violation	py-aipl/samples	runtime type cast at any-boundary	Py
Phase 17 structured conc.	Phase17_StructuredConc	py-aipl/samples	<code>scope { future ... }</code> auto-joining	Py
Session types / protocol traces	SessionTyped	py-aipl/samples-ai	runtime-checked typed session protocols (arg + reply types); order-only protocol traces; AIOS service registry	Py
AI integration (mock + real)	AIActor, AIChain, AIChainReal, AIHello, MultiProvider	py-aipl + samples-ai + abclcl/ai-samples	LLM-backed actors; multi-provider; chained pipeline	Py, OCaml
AI governance	Budgeted	py-aipl/samples-ai + abclcl/ai-samples	token budget, concurrency cap, fallback chain	Py, OCaml
AI cooperative pattern	CooperativeNowFuture{-jp,-jp-remote}, CooperativeSolve{-jp,Remote,Remote-jp}, Reviewer, Fanout, PriorityFanout	py-aipl/samples-ai + abclcl/ai-samples	Planner→Solver→Reviewer; fan-out aggregator; priority routing	Py, OCaml

Feature category	Sample(s)	Where	What it checks	Target
Remote actors	client, coordinator, verifier, RemoteCalcClient/Server	server, solver, reviewer_node*,	py-aipl/samples-remote + abclc/samples-remote + abclc/ai-samples	HTTP cross-machine sends; HMAC-signed coordination
Web / dashboard	web_calc, SiteGen	abclc + py-aipl/samples	embedded HTTP gateway; static-site generator	Py (SiteGen), OCaml (web_calc)
GUI / SDL2	Rotate{One,Three,Four}Lines{,Gui}, MultiLineSpin, Philosophers5Gui, BoundedBufferGui, DisasterReturnGui, LineDrawer, window	abclc	SDL2-backed GUI codegen via aipl2c	SDL2 (via aipl2c)
Python codegen target	BoundedBufferPy, Philosophers5Py, Rotate4LinesPy	abclc	aipl2c --python emits stand-alone Python	Py-gen
Xinu (embedded OS) target	BoundedBufferXinu, Philosophers5Xinu, Rotate4LinesXinu	abclc	aipl2c --xinu emits Xinu-flavoured C	Xinu
Pony codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --pony emits Pony source; class → actor , methods → be ; two-step _aipl_init decouples construction from init body	Pony
Erlang codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --erlang emits an .erl module; class → spawn + receive loop; fields → loop args with versioned variables on assign	Erlang
Go codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --go emits a single main.go ; class → struct + goroutine; methods → typed message structs + helper methods + type-switch dispatch	Go
Prolog codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --prolog emits a single SWI-Prolog .pl using library(thread) ; class → c_loop(Fields) thread + thread_get_message + Msg = m(Args) → body dispatch; expressions hoisted to goals (X is A + B , format(atom(S), ...))	Prolog

Feature category	Sample(s)	Where	What it checks	Target
LLVM codegen target	Hello, counter (verified)	abcl	<code>aipl2c --llvm</code> emits the default C with clang-specific <code>__attribute__((hot)) / ((cold))</code> on the actor / watchdog / print paths, plus a header banner listing <code>clang -O2 -pthread</code> (native), <code>clang -emit-llvm -S</code> (text IR), and <code>clang -emit-llvm -c + lli</code> (interpret) workflows	LLVM
OpenMP codegen target	Hello, counter (verified)	abcl	<code>aipl2c --openmp</code> emits C + <code><omp.h></code> ; the initial actor-spawn loop is converted to <code>#pragma omp parallel for schedule(dynamic)</code> and the message counter uses <code>#pragma omp atomic</code> . Build with <code>gcc-15 -fopenmp</code> or <code>clang -fopenmp</code> . pthreads still drive per-actor message loops (coexist with OpenMP)	OpenMP
Drone / simulation	<code>drone_simulator</code>	browser-abcl	obstacle-aware drone swarm with comm + view range	JS-B, JS-N
Trace / minimal repros	H, P, T*, LD*, MS, AA, PP, PH, line*, Philosophers5_{debug,trace}	abcl	reduced repro cases used during runtime / TLA+ / Spin model-checking	OCaml

Core language

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
method injection (<code>add_method/remove_method</code>)	✓	✓	✓	✓	×	×	×
now synchronous send	✓	✓	✓	✓	✓	✓	✓
future async send	✓	✓	✓	✓	✓	✓	✓
await future block	✓	✓	✓	✓	✓	✓	✓
send fire-and-forget	✓	✓	✓	✓	✓	✓	✓
become actor class swap	✓	✓	✓	✓	×	×	×
select selective receive ^f	✓	✓	✓	✓	✓	✓	×
top-level functions	✓	✓	✓	✓	×	×	✓
signatures / overloads	×	×	✓	✓	×	×	×
dynamic compile (<code>compile()</code>)	✓	✓	✓	✓	×	×	×
worker pool / <code>DynamicWorkerPool</code>	✓	✓	✓	✓	△	△	×
text file I/O (<code>read_file/write_file/append_file/file_exists</code>)	✓	✓	✓	✓	✓ ^a	✓ ^a	×
image I/O (<code>image_create/image_load/image_save/image_size/image_pixel/image_set_pixel</code>)	✓ ^b	✓ ^b	✓ ^c	✓ ^c	✓ ^{a,c}	✓ ^{a,c}	×

^a JS-B / JS-N の I/O ビルトインは host が Node の `fs` を注入したとき (`server.mjs` が自動で行う) のみ動作。ブラウザ内では "not available" を投げる。

^b Py-A / Py-I は Pillow 経由で PNG / JPEG / GIF / WebP を判定して読み書き。

^c OCaml / JS-O / JS-B / JS-N は純ランタイムの PPM (P6) バックエンド (RGBA メモリ表現、保存時はアルファ破棄)。PNG は外部ライブラリが必要。

^f Python `select { case ... }` 文は commit 9aaf371 (2026-05-18) で追加された (`grammar.lark`, `aipl_ast.SelectStmt`, `aipl_parser.Builder.select_*`, `aipl_interp.do_select`)。OCaml `parser.mly` の `select_cases / select_pat / select_timeout` と同じ shape / semantics。チャネルベースの `select_recv(...)` は Python では別ビルトインとして引き続き利用可能。

^g 29e / 29f は send-and-receive の 2 つの等価 shape。reply-on-send routing (commits 590e3aa OCaml / dd9d604 Python) のおかげで、`Worker.reply(r)` は受信側 mailbox に `reply(r)` メッセージとして配送される。(e) は通常メソッド定義 `method reply(r) {...}`、(f) は明示 `select { case reply(r) -> ... }`。C ランタイムは reply slot 自体無いため ×。サンプル: `abclc/ai-samples/SendReplyMethod.abcl` (OCaml cbec37c) と `src/python-aipl/samples-ai/_send_reply_demo/*.abcl`。

^h CE-10 effect type inference: inference walk (`aipl_inference._collect_effects_from_ast`) accumulates side-effect categories from every primitive call site, using the `BUILTIN_EFFECTS` table maintained in `aipl_typecheck.py` (Phase 12 system, 43 primitives covering `fs / ai+net / mut`)。InferenceResult.effects (inferred) と declared_effects (from `!{eff1, eff2}` clause) を render し、declared が inferred を被覆しないとき inferred but not declared を表示。2026-05-18 MAP-Elites AIPL_PyI_NextGen.aice で reviewer 0.70 で 1 位。

ⁱ DR-13 auto-scaling actor pool: hysteresis-driven 動的 actor プール。4 prims `pool_create(cls, min, max, target_qlen[, name]) / pool_pick / pool_size / pool_destroy` で構成。pool_pick 毎に平均 qlen を判定し > 2*target なら spawn (size < max)、< target/2 なら retire (size > min)。AIPL_DIST_ENABLE=1 でのみ動作。イベント `pool_created / pool_scale_up / pool_scale_down / pool_destroyed` は既存の structured-log NDJSON に流れる。サンプル: `src/python-aipl/samples/AutoScalingPool.abcl`。MAP-Elites reviewer 0.67 で distribution 軸 1 位。

^j CE-11 capability types: Phase-12 静的 effect system に乗った runtime capability tracking。スレッド毎に held caps の set (TLS) を持ち `AIPL_CAP_GRANT="fs,ai,net,mut"` で起動時 seed。5 prims: `grant_cap / revoke_cap / has_cap / current_caps / check_capability(name | names)`。Cap 名は `BUILTIN_EFFECTS` の effect tag と同一語彙 (`fs / ai / net / mut`)。Advisory 既定 (`cap_violation log`)、`AIPL_CAP_STRICT=1` で `CapabilityError` 強制 raise。サンプル: `src/python-aipl/samples/CapabilityTypes.abcl`。round-2 MAP-Elites reviewer avg 0.430 で type 軸 1 位 (top elite I0012 で 0.73)。

^k DR-10 CRDT actor state: G-Counter (grow-only, per-replica max merge) / OR-Set (observed-remove, UUID add-tags) / LWW-Register (last-writer-wins, ts + replica-id tiebreak) の 3 古典的 CRDT を plain dict として実装し `save_actor_state / restore_actor_state round-trip` を温存。15 prims (各型 4-6 個 + `crdt_replicate(name, value)` hook)。Replica id は `AIPL_DIST_REPLICA_ID` (default hostname)。crdt_replicate は deployment-layer replicator 向けのイベントを log、peer fan-out 自体は将来の multi-region driver (DR-12) に分離。サン

ル: `src/python-aipl/samples/CRDTState.abcl`. round-2 MAP-Elites reviewer avg 0.410 で distribution 軸 2 位。

¹ DR-12 multi-region failover: DR-1 route table の region-aware 拡張。各ノードは `AIPL_REGION=<name>` (default local) を持ち、ルートテーブルは `AIPL_ROUTE_REGION_<name>="Actor:tag,..."` で region 毎に持つ。Primary region で actor が見つからないとき、`AIPL_REGION_FAILOVER="us-east-1,eu-west-1,ap-1"` を順に walk して該当 region を探す。region_failover (cross-region hit) / region_failover_failed (chain 終了) を log。5 prims: `current_region`, `region_chain`, `route_for_region(actor[, region])`, `failover_region(actor[, primary])`, `regions_available`。サンプル: `src/python-aipl/samples/MultiRegionFailover.abcl`. round-2 MAP-Elites reviewer avg 0.405 で distribution 軸 3 位 (top elite I0012 で 0.70)。

° CE-12 refinement unification: HM の unify は従来 refinement 述語を drop して post-hoc `_check_refinements` で end-of-pass に Z3 sweep を回していた。CE-12 は両側 TRefined の unify 時、`AIPL_REFINE_UNIFY=1` 設定下で Z3 に $P_{\text{actual}} \wedge \neg P_{\text{expected}}$ の UNSAT を問い (subset 関係 $P_{\text{actual}} \Rightarrow P_{\text{expected}}$ 成立)、不成立なら `UnifyError` を直ちに raise (post-hoc を待たず call site で early flag)。Env var 未設定なら従来挙動と互換 (drop refinement)。サンプル: `src/python-aipl/samples/RefinementUnification.abcl`. round-2 MAP-Elites reviewer avg 0.383 で 8 機能中 8 位 (Py-I 8 機能全完了)。

ⁿ CE-13 record width subtyping: `aipl_inference.unify` の record arm は従来 TRecord のフィールド集合完全一致を要求していたが、CE-13 で intersection-based に緩和。両側に出現するフィールドのみ pairwise unify (depth subtyping 維持)、片側のみのフィールドは無視 (width subtyping)、完全 disjoint なら従来通り fail。Unannotated record param の principal-type 挙動は維持 (アクセスされた union がパラメータ TVar の制約として累積)。サンプル: `src/python-aipl/samples/RecordSubtyping.abcl`. round-2 MAP-Elites reviewer avg 0.400 で残り type 軸候補のうち 1 位 (CE-11 以降)。

^m DR-11 saga orchestration: 補償アクション・オーケストレーションを新規 top-level 文として導入。grammar.lark に `saga_stmt: "saga" "{" saga_step+ "}"` と `saga_step: "step" saga_block "compensate" saga_block` を追加 (`saga_block` は明示的非終端で transformer が 2 つの Block 子を実際に取得)。`aipl_ast.SagaStmt` / `SagaStep` + `aipl_interp.do_saga`。forward は各 `step.body` を順実行、失敗時は completed step の `compensate` を LIFO で実行してから例外を re-raise (actor 標準 error path + IQ quarantine が発火)。8 events: `saga_started` / `saga_step_complete` / `saga_step_failed` / `saga_compensated` / `saga_compensate_failed` / `saga_finished` / `saga_aborted`。サンプル: `src/python-aipl/samples/SagaOrchestration.abcl`. round-2 MAP-Elites reviewer avg 0.406 で残り分散候補の中で 1 位。

Type system

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
type inference	△trace	✓HM	✓HM	✓HM	✓flow	✓flow	✓HM+spec
type annotations	✓	△ignored	✓	✓	×	×	×
records {a:int, b:str}	✓	✓	✓	✓	×	×	△type only
tuples (1, "a")	✓	✓	✓	✓	×	×	×
arrays (typed, multi-dim)	✓	✓	✓	✓	✓	✓	△
generics on functions	×	✓Forall	✓Forall	✓	×	×	×

Phase 11+ advanced features

Python (inferred) runs HM only, so the annotation-driven Phase 11+ static checks that Python (annotated) performs are not re-implemented. Programs still execute at runtime via the shared interpreter, but the static guarantees from Phase 12 / 14 / 15 / 16 are lost. Phase 13 (channels), Phase 17 (scope), and the session-type / protocol-trace / AIOS-registry families are runtime features, so they remain available in Py-I.

Session types here are *runtime-checked* and explicitly separate from HM inference: they observe values flowing over the wire at runtime and validate them against a declared protocol spec ("`actor.method(arg_t1, ...) ! ret_type -> ...`"). Violations are recorded into `session_events()` and counted; the program does not abort.

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
channels (CSP-style) —Phase 13	✓	✓	×	×	×	×	×
linear types / use-after-move —Phase 14	✓	×	×	×	×	×	×

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
owned / pub fields —Phase 15	✓	×	×	×	×	×	×
effects $\{\text{fs}, \text{ai}, \text{net}, \text{mut}\}$ —Phase 12	✓	×	×	×	×	×	×
structured concurrency scope —Phase 17	✓	✓	×	×	×	×	×
transient cast at any-boundary —Phase 16	✓	×	×	×	×	×	×
session types (runtime-checked typed protocols)	✓	✓	×	×	×	×	×
protocol traces (order-only)	✓	✓	×	×	×	×	×
AIOS service registry (alias-resolved send)	✓	✓	×	×	×	×	×

Phase C–E2 type inference (Python annotated only, added 2026-05-17/18)

Phase 11–17 の checker layer に加え、Python (annotated) ランタイムは **constraint-based Hindley–Milner + Z3 refinement** 推論系を別モジュール `aipl_inference.py` (~1255 LOC) として持つ。Py-I (`python-aipl-inferred/`) は依然 trace-based HM のままで、これらの新 phase は取り込まれていない。

ID	Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
CE-1	Phase C: constraint-based HM + Z3 refinement (Int)	✓	×	△**	△**	×	×	×
CE-2	Phase D-1: cross-class inference	✓	×	✓#	✓#	×	×	×
CE-3	Phase E- α : where 句 in AIPL grammar	✓	×	✓*	✓*	×	×	×
CE-4	Phase E- β : actor field 共有 (method 横断)	✓	×	✓#	✓#	×	×	×
CE-5	Phase E- γ : record structural typing	✓	×	✓##	✓##	×	×	×
CE-6	Phase E- γ -R: Real / Rat refinement (Z3 Real)	✓	×	△**	△**	×	×	×
CE-7	Phase E-2: typeck × inference 統合 CLI <code>--check</code>	✓	×	✓\$	✓\$	×	×	×
CE-8	<code>--infer</code> standalone CLI (Phase D-4)	✓	×	✓\$	✓\$	×	×	×
CE-9	refinement vacuously-false detection (declaration-time, Z3 unsat)	✓	×	△**	△**	×	×	×
CE-10	effect type inference (<code>{ai, fs, net, mut}</code> rows in inferred method types) ^h	×	✓	✓	✓	×	×	×
CE-11	capability types (<code>grant_cap / revoke_cap / has_cap / check_capability</code> , strict-mode via <code>AIPL_CAP_STRICT=1</code>) ^j	×	✓	✓	✓	×	×	×
CE-13	record width subtyping (TRecord unify → intersection-based) ⁿ	×	✓	✓	✓	×	×	×
CE-12	refinement unification (early Z3 subset check in unify under <code>AIPL_REFINE_UNIFY=1</code>) ^o	×	✓	△	△	×	×	×

サンプル: `aice-pi-evolution/experiments/2026-05-17_aipl_v2_type_inference/samples/feature_{a..g}/` (7 feature × 3 = 21 demo + 27/27 unit tests).

* CE-3: OCaml は Phase O-2.a (2026-05-18) で parsing 達成. `lexer.mll` に `where/and/or/not` キーワード, `parser.mly` に `type_expr WHERE refine_or` 規則, `ast.ml` に `TyERefined of type_expr * refine_pred` を追加. サンプル: `abclc/WhereClause.abcl`.

** CE-1 / CE-6 / CE-9: OCaml は Phase O-2.b (2026-05-18) で SMT-LIB 2 + z3 CLI バックエンドを実装, Phase O-2.c (同日) で Real 理論にも対応. `src/refinement.ml` が `Ast.refine_pred` を `smt_sort` (Int / Real) パラメータ付きで SMT-LIB 2 にレンダし `z3 -in -t:5000` に流す. `AIPL_REFINE_CHECK=1` で typecheck pass が vacuously-false な refinement に [refine warning] を吐く. z3 不在時は Deferred で graceful fallback. サンプル: `abclc/WhereVacuous.abcl` (Int), `abclc/WhereVacuousReal.abcl` (Real).

CE-2 / CE-4: OCaml は Phase O-2.d (2026-05-18) で cross-class 推論を強化. `preinfer_all_classes` が declared return-type を尊重し, `Now / Future` expression が `class_method_schemes` 越しに actual ret 型を返すよう変更. `TypedVarDecl` の unify 失敗を `Type_error` 化. actor field の cross-method 一貫性 (Int/String mix 検出) も自動取得.

CE-5: OCaml は Phase O-2.e (2026-05-18) で record structural typing を完成. `Types.unify` の TRecord ケースは既に label-sort + 再帰 unify を実装していたが, `preinfer_all_classes` が method parameter annotation を無視していたため cross-class 呼出しで shape 不一致が silent だった. parameter annotation を尊重するよう修正後, `method f(p:{a:int,b:int})` に shape 不一致の record literal を渡すと count / label / field-type の全てで type error. サンプル: `abclc/RecordStructural.abcl`.

\$ CE-7 / CE-8: OCaml は Phase O-2.f (2026-05-18) で CLI 統合. `repl_thread.exe` に `--type-check FILE / --infer FILE / --check FILE / --strict` を追加. `--check` は Python-style の section banner 付き出力, `--strict` で issue 時に exit 3. Z3 refinement は `AIPL_REFINE_CHECK=1` を併用すると [refine warning] vacuously-false ... が stderr に発火.

OCaml ランタイムも HM 推論を持つが (#12 で ✓), その他の refinement 機能 (CE-1, CE-6, CE-9) と record structural typing (CE-5) は今後の課題.

AIPL v2 Distributed runtime (**aipl_dist**, added 2026-05-18)

進化計算 (MAP-Elites 8 seed × 30 gen × 2 run) で発見された分散ランタイム設計 (I0003 balanced / I0023 hang-resilience / I0036 Erlang OTP の 3 候補) を **aipl_dist.py** (591 LOC) として実装。AIPL 言語仕様は不変, 既存サンプルは無改変で同じ挙動。 **AIPL_DIST_ENABLE=1** で全機能 **opt-in**。

ID	Feature (env var)	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
DR-1	I-1 env_var_routing (AIPL_ROUTE)	✓	×	✓	✓	×	×	×
DR-2	I-2 structured_log (AIPL_DIST_LOG_FILE)	✓	×	✓	✓	×	×	×
DR-3	I-3 token_budget_aware (AIPL_DIST_RPM / TPM)	✓	×	✓	✓	×	×	×
DR-4	I-4 checkpoint_and_resume (AIPL_DIST_CHECKPOINT_DIR)	✓	×	✓	✓	×	×	×
DR-5	IQ quarantine_and_skip (AIPL_DIST_QUARANTINE_TTL)	✓	×	✓	✓	×	×	×
DR-6	IM-1 quorum_replicate (AIPL_DIST_QUORUM_PROVIDERS)	✓	×	✓	✓	×	×	×
DR-7	IM-2 subtree_quarantine (AIPL_DIST_SUBTREE_QUARANTINE)	✓	×	✓	✓	×	×	×
DR-8	spawn-tree tracking (TLS-auto parent)	✓	×	✓	✓	×	×	×
DR-9	runtime hooks (interp / actor / call_ai) — opt-in	✓	×	✓	✓	×	×	×
DR-13	auto-scaling actor pool (pool_create / pool_pick / pool_size / pool_destroy with hysteresis) ⁱ	×	✓	✓	✓	×	×	×
DR-10	CRDT actor state (G-Counter / OR-Set / LWW-Register; 15 prims + crdt_replicate hook) ^k	×	✓	✓	✓	×	×	×
DR-12	multi-region failover (per-region route + failover_region chain walk; 5 prims) ^l	×	✓	✓	✓	×	×	×
DR-11	saga orchestration (saga { step {...} compensate {...} ... } statement, LIFO compensate-on-failure) ^m	×	✓	✓	✓	×	×	×

実装位置: src/python-aipl/aipl_dist.py (591 LOC, 新規) + aipl_interp.py (+27 行 spawn hook) + aipl_runtime.py (+13 行 error hook) + aipl_ai.py (+13 行 budget hook).

サンプル: aice-pi-evolution/experiments/2026-05-17_aipl_v2_type_inference/IMPL_I0003_MVP/samples/ (8 feature × 3 = 24 demo + 27/27 unit tests).

設計探索の出自: 同 dir の AIPL_v2_Distributed.{aice,ga,json,schema,json} を MAP-Elites で 24 min × 2 run。詳細は IMPL_I0003_MVP/IMPL_DESIGN.md ~ IMPL_I0036_REPORT.md 参照。

OCaml は Phase O-1 + O-1.5 で全 9 機能 ✓ 達成。DR-3/DR-6 は Ai.call_gemini へ自動 wire 済, DR-8 は per-thread current_actor TLS で parent を自動取得。

Phase O 完全完了サマリ (2026-05-18)

領域	Py-A	OCaml	JS-O	LOC (実装)
Distributed (DR-1~9)	✓9/9	✓9/9	✓9/9	~720
Type Inference (CE-1~9)	✓9/9	✓9/9	✓9/9	~585
合計	18/18	18/18	18/18	~1305

OCaml AIPL は Python (annotated) と ** 完全機能同等 ** に到達. 実装規模は当初推定 ~1925 LOC の **68%** で済んだ. 詳細は docs/OCAML_PORT_ROADMAP.md.

Phase O-3 —Send/Select 相互作用整合 (2026-05-18)

OCaml と Python (型推論) で send / now / future / select / reply の細部の挙動を完全対称化.

Fix	内容	OCaml	Python (型推論)	LOC
(1)	reply(v) を送信側 mailbox に配送	✓590e3aa	✓dd9d604	+25
(2)	new X() in-method actor_table key 整合	✓590e3aa	N/A (元から OK)	+5
(3)	script-mode actor quiescence	✓590e3aa	✓dd9d604	+60
(4)	Python select { case ... } 構文追加	(元から ✓)	✓9aaf371	+134

両ランタイムで成立する 2 つの receive shape:

- (e) user-defined method reply(r) { ... } —Erlang プロセス風, actor loop が自動 dispatch. サンプル: abclc/ai-samples/SendReplyMethod.abcl (OCaml cbec37c) / samples-ai/_send_reply_demo (Python).
- (f) select { case reply(r) -> ... } —OTP / Erlang receive 風. サンプル: _ai_call_styles/send_select_style.abcl (OCaml) / samples-ai/_send_reply_demo/send_select_style.abcl (Python).

両ランタイムに 4 形態 (now / future+await / send+reply-method / send+select) の ai_call デモが揃った. 実 OpenAI provider + mock 両方で動作確認済..aice 仕様書: aice-pi-evolution/experiment

Networking, AI, infrastructure

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
remote actors (HTTP)	✓	✓	✓ ^d	✓ ^d	×	△ _{srv} only	✓ _{via} OCaml
WebSocket	✓ _{ws_listen} builtins (web-sock-ets lib)	✓ _{send/close} A)	✓ _{in} (ver-ified 101)	✓ _{via} OCaml	✓ _{browser} native	✓ _{ws?sid=abcl} + /api/broadcast (ws lib)	✓ _{abcl_ws_runtime.c} + lib-sock-ets
AI integration (ai_call)	✓ _{stream/img}	✓	✓	✓	✓ _{mock}	✓ _{mock}	✓
provider arg (ai_call([1 2 3,] prompt)) —1=gemini, 2=anthropic, 3=openai	✓	✓	✓	✓	✓	✓	✓ _{(via} OCaml)
now actor.m(...) + ai_call inside method (blocking)	✓	✓	✓	✓	✓	✓	×
future actor.m(...) + await(f) + ai_call	✓	✓	✓	✓	✓	✓	×
send + callback pattern with ai_call	✓	✓	✓	✓	✓	✓	×
send + user-defined reply(r) method ^g	✓	✓	✓	✓	✓	✓	×
send + select { case reply(r) -> ... } ^g	✓	✓	✓	✓	✓	✓	×
AI governance (budget/concurrent/fallback)	✓	✓	✓	✓	×	×	✓
HMAC-signed remote send	✓	✓	✓ ^e	✓ ^e	×	×	✓
persistent actor state	✓	✓	×	×	×	×	×
live dashboard (SSE)	✓	✓	△ _{polling}	△	×	×	✓
/api/typecheck JSON endpoint	×	×	✓	✓	×	✓	—
/api/run JSON endpoint	×	×	×	×	×	✓	—

Actor concurrency model

Every AIPL implementation runs each actor as a separate concurrent unit, but the kind of unit differs by 3–4 orders of magnitude in weight. This affects how many actors a program can usefully spawn, how messages interleave, and whether blocking I/O in one actor stalls the whole runtime.

Runtime / codegen tar- get	Model	Per-actor unit	Source pointer
Python (annotated)	1:1 OS thread	threading.Thread(daemon=True)	aipl_runtime.py:92
Python (inferred)	1:1 OS thread	(inherits from python-aipl)	(re-uses interp)
OCaml	1:1 OS thread	Thread.create	eval_thread.ml:~1168
JS-OCaml (server)	1:1 OS thread	(= OCaml backend threads)	(= OCaml)
JS-Browser (browser-abcl)	cooperative event loop	setTimeout-driven mailbox drain	runtime.js:~233
JS-Node (node-aipl-server)	cooperative event loop	(= browser-abcl runtime)	re-export
C (default)	1:1 OS thread	pthread_create	c_translator.ml:~721,790

Runtime / codegen target	Model	Per-actor unit	Source pointer
C + SDL2	1:1 OS thread	pthread_create	(same as default C)
C + Xinu	1:1 OS process (kernel-level)	Xinu create() + ready()	c_translator.ml:~1077
C → Python codegen	1:1 OS thread	threading.Thread (o._spawn())	c_translator.ml:~1847
C → Pony codegen	M:N lightweight (runtime-scheduled)	Pony actor (work-stealing)	implicit in generated actor
C → Erlang codegen	M:N lightweight (BEAM)	BEAM process (spawn(fun()->...end))	c_translator.ml:~2413
C → Go codegen	M:N lightweight (Go runtime)	goroutine (go c.run())	c_translator.ml:~2728
C → Prolog codegen	1:1 OS thread	SWI thread (thread_create/3)	c_translator.ml:~3060

Three concurrency tiers

- **1:1 OS thread / process** (Python, OCaml, C, C-SDL2, Xinu, C→Python, C→Prolog): each actor is a kernel-scheduled thread. Simple mental model; blocking I/O in one actor doesn't stall others. Scales to ~100–1000 actors before kernel overhead dominates.
- **M:N lightweight** (Pony, Erlang, Go): actors are scheduled by the language runtime onto a small pool of OS threads. Scales to $\sim 10^5$ – 10^7 actors. Blocking syscalls in one actor may stall its scheduler unless the runtime intercepts them (Erlang's BEAM intercepts; Go's runtime intercepts via `netpoll`; Pony's runtime requires non-blocking idioms).
- **Cooperative event loop** (browser-abcl, node-aipl-server): single OS thread; mailboxes are drained turn-by-turn through `setTimeout(0)` re-entry into the event loop. No real parallelism, but message-level interleaving is preserved. A long synchronous computation in one actor *will* stall all others — by design, since this is the only concurrency model the browser DOM allows.

Cross-cutting implication

The AIPL programming model (`send` / `now` / `future` / `await`) is preserved across all three tiers — the choice of target picks a point on the cost/scale curve without changing the program text. The same `.abcl` file:

- runs ~1000 actors fine on Python / OCaml / C (OS-thread tier);
- scales to millions of actors on Erlang / Go / Pony codegen;
- runs in a browser sandbox via JS-Browser at the cost of single-threaded execution.

C-version-specific codegen targets

Target	C
C + pthread standalone binary	✓
C + SDL2 GUI binary (1178-line runtime)	✓
Xinu embedded-OS target (<code>--xinu</code>)	✓
Python target (<code>--python</code>)	✓
Pony target (<code>--pony</code>) —class→actor, methods→be, two-step <code>_aipl_init</code>	✓
Erlang target (<code>--erlang</code>) —class→spawn+receive loop, fields→versioned loop args	✓
Go target (<code>--go</code>) —class→struct+goroutine, methods→typed message structs + type-switch dispatch	✓
Prolog target (<code>--prolog</code>) —class→SWI thread + receive loop, expressions hoisted to goals (<code>is/2</code> , <code>format(atom(...))</code>)	✓

Key observations

Python (annotated) is the most feature-complete runtime

Of the 30 surveyed features, Python (annotated) ticks roughly 28. It owns the full Phase 11–17 stack (channels, linear types, owned fields, effects, structured concurrency, transient casts), method injection, dynamic compile, persistent actor state, full AI integration (streaming + multimodal images), and Pillow-backed image I/O (PNG / JPEG / GIF / WebP).

Python (inferred) — Python’s HM-typed twin

Runtime features match Python (annotated) exactly because the interpreter is shared. The trade is on the static side: the Phase 11+ annotation-driven checks are replaced with a Hindley-Milner inference pass. Inferred types cross-verified against OCaml: on shared samples (`Hello.abcl`, `counter.abcl`) the two runtimes produce identical type schemes.

OCaml — feature parity with Python on the core

After the OCaml feature catch-up, the runtime supports **method injection** (`add_method/remove_method`), **dynamic compile** (`compile()` + `spawn()`), **top-level functions** with **generics** (`function id(x: T) -> T`), **type annotations**, **records & tuples**, **sized arrays** (`var x[N] [M]`), and **text/image file I/O** (PPM backend for images). Phase 11+ effect / linear / owned / transient features still only exist as `.abcl` design-document samples — not statically enforced. Solid **become** / **select** / WebSocket (built into `web_gateway.ml`) and HTTP remote actor calls (`send` / `now` / `future`, error-tolerant) with **HMAC-signed traffic** (`ABCL_REMOTE_SECRET` 環境変数, 純 OCaml HMAC-SHA256 in `src/hmac_sha256.ml`, Python / C 版とワイヤ互換).

^d OCaml の remote client (`src/remote_client.ml`) は `send` (fire-and-forget) / `now` (同期 reply) / `future` (並列 + await) の 3 形式すべてをサポート。ECONNREFUSED / timeout は stderr に記録され、呼び出し側のアクターは生存する。エンドツーエンド検証は `src/test_remote_actor.sh` (9/9)。JS-O は OCaml バックエンド経由でこれを継承する。

^e 純 OCaml HMAC-SHA256 (`src/hmac_sha256.ml`, RFC 6234 / 4231 テストベクタで verify) を sender と receiver の両側で使う。`ABCL_REMOTE_SECRET` が両端でセットされたとき、全リクエストに `X-ABCL-Sig: <hex>` が付与される。`web_gateway.ml` の `verify_hmac_or_reject` は不一致なら 401 を返す。`python-aipl/aipl_remote.py` と C ランタイムにワイヤ互換。以前は受信側が `openssl dgst` をサブプロセス起動していたが、インプロセスの純 OCaml 版に置換 (リクエスト毎の fork 排除)。

JS-OCaml $\approx$ OCaml

A thin HTTP client (`src/app.js` etc.) over the OCaml backend exposed by `web_gateway.ml`. Every OCaml-side feature — method injection, dynamic compile, top-level functions, generics, type annotations, file & image I/O — is reachable via `/api/repl` from a browser or curl. The OCaml-specific differentiator is the `/api/typecheck` JSON endpoint.

JS-Browser — minimal in-browser engine

Basic actor model plus flow-sensitive type inference. Recently picked up **sized arrays** (`var x[N]`) and **text & image file I/O** (host-injected `fs`; throws in browsers). No **become**, no channels, no remote, no method injection. AI integration exists but is mock-only.

JS-Node $\approx$ JS-Browser + `fs`

Wraps the browser-abcl runtime inside a Node.js HTTP server (`src/node-aipl-server/server.mjs`); auto-injects Node’s `fs` so the text & image I/O builtins are functional. Exposes `/api/typecheck` and `/api/run` for headless use. Same flow-sensitive type inference as browser-abcl.

C codegen — broadest target diversity

`become` and `select` remain codegen-unsupported (emit `/* unsupported */`), but the codegen inherits OCaml’s HMAC, SSE, and AI governance, and offers **9 backends** from one front-end:

- C + pthread (default)
- C + LLVM / clang (`--llvm: adds __attribute__((hot/cold))`) + headers for clang `-emit-llvm`)
- C + OpenMP (`--openmp: #pragma omp parallel for spawn loop + #pragma omp atomic counter`)
- C + SDL2 (GUI demos)

- Xinu / Python / Pony / Erlang / Go / SWI-Prolog

Fields, parameters, and locals are specialised to native C types via HM inference + per-class field-type registry.

Method injection — now in three runtime families

Both Python variants and the OCaml runtime (with JS-OCaml piggy-backing on it) implement mutable method dispatch tables with runtime `add_method` / `remove_method` / `methods_of`. `browser-abcl` and `node-aipl-server` still lack even `become` and rely on the unchanging method table baked at parse time.

Type inference cross-verification

For shared samples (`abclc/Hello.abcl`, `abclc/counter.abcl`), the three HM-based runtimes (**OCaml**, **Python-inferred**, **C**) produce identical inferred types:

```
Hello:  count : float
        init  : (int) -> unit
        greet : ()  -> unit
        inc   : ()  -> unit

Counter: count : float
        inc   : ()  -> unit
        dec   : (float) -> unit    <- monomorphized via call site
```

The flow-sensitive variants (**JS-Browser**, **JS-Node**) reach the same answers on shared samples within the limits of their algorithm —field types match, but method signature schemes are not produced the same way.

Phase 11+ implementation gap

The full Phase 11–17 stack is implemented only in Python (annotated). This represents AIPL’s language-evolution frontier; Python (annotated) serves as the research runtime. Python (inferred) keeps the runtime side but trades the static checks for HM-style polymorphic inference — a different research axis (declarative typing vs. annotated capabilities).

Post-Phase-11 (added 2026-05-17/18)

領域	Py-A 達成	他ランタイム	関連 ID
Phase C–E2 型推論 (HM + Z3 refinement)	9/9	0/9	CE-1 ~ CE-9
AIPL v2 Distributed (<code>aipl_dist</code>)	9/9	OCaml 9/9 (= JS-O)	DR-1 ~ DR-9

Py-A 単独で進化計算由来の 18 機能が opt-in 利用可能 (`AIPL_DIST_ENABLE=1` または `--check` / `--infer` で発火, デフォルトは従来挙動と同一)。OCaml 側への移植 path:

- 型推論側: `--infer` の HM-with-refinement を OCaml HM (`src/infer.ml`) に統合すれば CE-1, CE-2, CE-4 は取り込み可能. `where` 句 (CE-3) は OCaml lexer/parser 修正が必要.Z3 backend (CE-1, CE-6, CE-9) は別プロセス via JSON が現実的.
- ランタイム側: DR-1 / DR-2 / DR-4 / DR-5 / DR-8 / DR-9 は OCaml `eval_thread.ml` + `aipl_remote.ml` への hook 追加 (~200 LOC) で実装可能.DR-3 / DR-6 は AI client 層, DR-7 は spawn-tree registry が必要.

Generated 2026-05-14, updated 2026-05-18. Cumulative through: OCaml feature catch-up (records / tuples / dynamic compile / method injection / top-level functions / generics / type annotations / sized arrays / text & image I/O); C-codegen LLVM / OpenMP targets; python-aipl-inferred (commit 270f291) and node-aipl-server (commit adeb0b6); **Phase C–E2 constraint-based HM + Z3 refinement** (aipl_inference.py, 1255 LOC; sections CE-1..CE-9); **AIPL v2 Distributed runtime** (aipl_dist.py, 591 LOC; sections DR-1..DR-9), discovered by MAP-Elites GA + MVP-implemented for I0003 / I0023 / I0036 winners.

Phase O complete (2026-05-18): OCaml AIPL port of all 18 CE-* + DR-* features (~1305 LOC actual vs ~1925 estimated). src/aipl_dist.ml, src/refinement.ml, and incremental hooks into infer.ml / eval_thread.ml / ai.ml / repl_thread.ml. JS-OCaml inherits all features through the OCaml backend.

For source pointers, run `grep` against the files listed in each runtime’s source column.

*Last regenerated: 2026-05-18 (after Py-I → **OCaml port of all 8 next-gen features**: CE-10/CE-11/CE-12/CE-13 + DR-10/DR-11/DR-12/DR-13. CE-12 OCaml is —Z3-subset API shipped but the unify-side early-flag hook is deferred since OCaml lowers TyERefined to base types before unification).*